

Volatility Surface

J.P. Morgan



Overview

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The Black-Scholes formula

Underlying asset: non-dividend paying stock or index with non-negative price process

$$\frac{dS_t}{S_t} = \sigma dW_t + r dt$$

Call option on S with strike K and expiry T pays $(S_T - K)_+$ at time T

The Black-Scholes option price depends on the contract variables (strike and maturity)

(K,T), on the present value of the underlying S(t) and on the short rate r

$$C_{BS}(K, T) = S_t N(d_1) - Ke^{-r(T-t)} N(d_2)$$

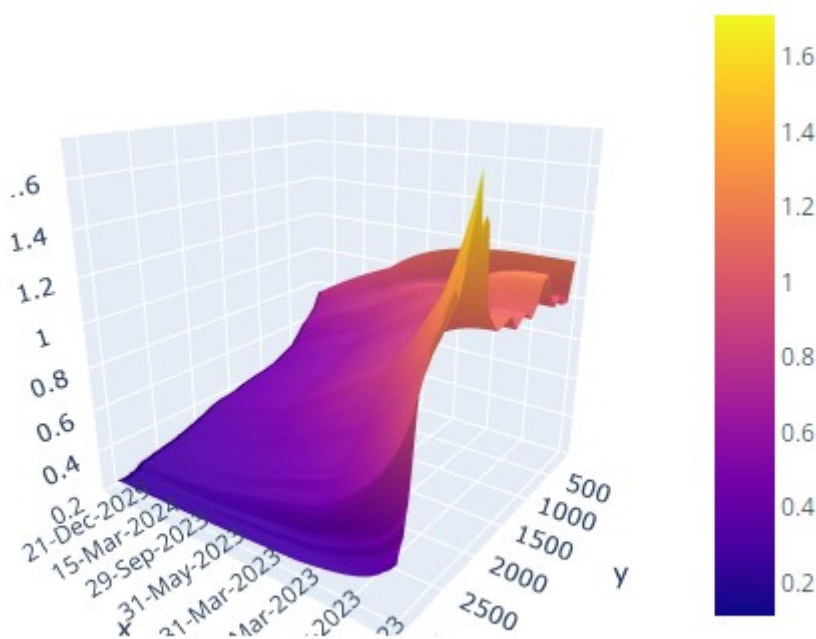
where

$$d_{1,2} = \frac{1}{\sigma \sqrt{T-t}} \left(\log \frac{K}{S_t} \pm \frac{1}{2} \sigma^2 (T-t) \right)$$

Volatility Surface

Off course real volatility is not constant.

- We observe that volatility as function of strike for a constant expiry is either symmetric (volatility smile, observed in FX) or downward sloping (volatility skew, observed in equities)
- Given prices of vanilla options for different strikes and maturities, one can invert black scholes formula to obtain implied volatility surface $\Sigma(K, T)$



SPX Volatility Surface as of 24-Feb-2023

- Why do we observe skew?
 1. Volatility depends on spot. There is empirical evidence for this. During periods of market crash, when spot drops -> volatility goes up. So maybe there is implicit relationship between spot and volatility.
 - Same intuition tells us that functional relationship must be inversely correlated
 - When stock and volatility are modelled as two separate process, their correlation acts as skew driver.
 - Capture this dependence via explicit function $\sigma_t = \sigma(t, S)$
 - This is local volatility model
 2. Volatility is itself a stochastic process.
 - This gives rise to stochastic volatility models.
 - We will not discuss these today.
 - Local volatility is simplest stochastic volatility model in which variance is perfectly correlated with stock

Local Volatility: Risk Neutral Density

- Let's work with undiscounted call prices
- $C(K, T) = E[(S_T - K)^+]$, taking 1st derivative with respect to K ,
- $\frac{\partial C}{\partial K} = -E[\theta(S_T - K)] = -P(S_T > K)$
- $-\frac{\partial C}{\partial K} = P(S_T > K) = 1 - P(S_T < K) = 1 - F(K)$ where F is cdf of stock price (1)
- Now let's take second derivative of $C(K, T) = E[(S_T - K)^+]$ with respect to K
- $\frac{\partial^2 C}{\partial K^2} = E[\delta(S_T - K)] = p(K)$ where p is risk neutral pdf of stock price (2)
- These equations are interesting. We have derived pdf and cdf of risk neutral stock process.
- And this is model independent. We haven't made any assumption yet!
- This is also known as Breeden Litzenberger formula

Local Volatility: Risk Neutral Density

- Quick applications of risk neutral density

1. Assuming deterministic rates, any derivative contract can be priced as

$$FV = e^{-rT} E^Q[\text{payoff}] = e^{-rT} \int p(S_T) \text{Payoff}(S_T) dS_T$$

2. We can use CDF of stock price for monte carlo simulations

- Generate uniform random number $[0,1]$
- Since cdf is always bound between these two number, take inverse of CDF at this random number value.
- This will generate S_T and hence payoff at maturity
- Take average over multiple randoms to calculate expected value of payoff

Local Volatility: Dupire equation

- Given observed prices of call and put option, how tightly can we constraint σ_t ?
- Let's start with call price directly and use Ito's Lemma on it
- Assume stock process is $dS_t = (r - q)S_t dt + \sigma_t S_t dW_t$
- $$d(S_T - K)^+ = \frac{d(S_T - K)^+}{dS_T} * dS_T + \frac{1}{2} \frac{\partial^2 (S_T - K)^+}{\partial S_T^2} * dS_T * dS_T$$
- $$d(S_T - K)^+ = \theta(S_T - K) * ((r - q)S_T dt + \sigma_t S_t dW_t) + \frac{1}{2} \delta(S_t - K) * \sigma_t^2 S_T^2 dt \quad (3)$$
- Where θ is heaviside function and δ is dirac delta function

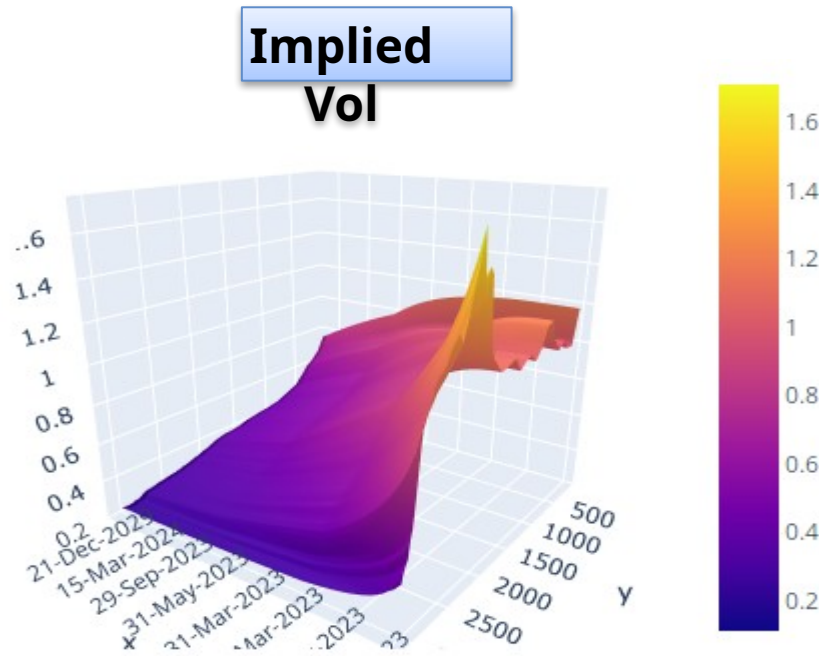
Local Volatility: Dupire equation

- $C(K, T) = E[(S_T - K)^+] = E[(S_T - K) * \theta(S_T - K)] = E[S_T * \theta(S_T - K)] - K * E[\theta(S_T - K)]$
- $C(K, T) = E[S_T * \theta(S_T - K)] + K * \frac{\partial C}{\partial K}$
- $E[S_T * \theta(S_T - K)] = C - K \frac{\partial C}{\partial K}$
- Now taking expectation of 3 from last slide,
- $E[d(S_T - K)^+] = E[\theta(S_T - K) * ((r - q)S_T dt + \sigma_t S_t dW_t) + \frac{1}{2} \delta(S_t - K) * \sigma_t^2 S_t^2 dt]$
- Consider LHS first. $E[d(S_T - K)^+] = d(E[(S_T - K)^+]) = dC = \frac{dC}{dT} * dT$. If you are not convinced about interchanging expectation against differential, use linearity of expectation or direct application of Leibniz integral formula.
- RHS is $(r - q) * (C - K \frac{\partial C}{\partial K}) dT + E[\frac{1}{2} \delta(S_t - K) * \sigma_t^2 S_t^2 dt]$
- $= (r - q) * (C - K \frac{\partial C}{\partial K}) dT + \frac{1}{2} K^2 E[\delta(S_t - K) * \sigma_t^2] dT$ we set $S = K$ because of delta function
- Final setup reads as $\frac{dC}{dT} = (r - q) * (C - K \frac{\partial C}{\partial K}) + \frac{1}{2} K^2 E[\delta(S_t - K) * \sigma_t^2]$

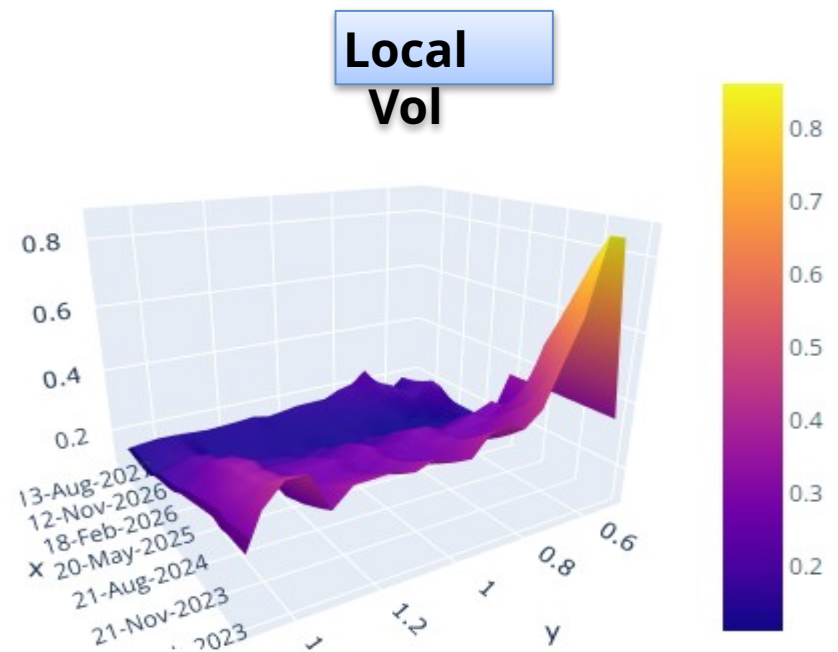
Local Volatility: Dupire equation

- From last page, $\frac{dC}{dT} = (r - q) * (C - K \frac{\partial C}{\partial K}) + \frac{1}{2} K^2 E[\delta(S_t - K) * \sigma_t^2]$
- Rearranging, $E[\delta(S_t - K) * \sigma_t^2] = \frac{\frac{dC}{dT} - (r - q) * (C - K \frac{\partial C}{\partial K})}{\frac{K^2}{2}}$
- Dividing LHS by $E[\delta(S_t - K)]$ and RHS by $\frac{\partial^2 C}{\partial K^2}$, we get
- $\frac{E[\delta(S_t - K) * \sigma_t^2]}{E[\delta(S_t - K)]} = E[\sigma_T^2 | S_T = K] = \frac{\frac{dC}{dT} - (r - q) * (C - K \frac{\partial C}{\partial K})}{\frac{K^2}{2} \frac{\partial^2 C}{\partial K^2}} = 2 \frac{\frac{dC}{dT} - (r - q) * (C - K \frac{\partial C}{\partial K})}{K^2 \frac{\partial^2 C}{\partial K^2}}$
- Let's put back interest rates now.
- $E[\sigma_T^2 | S_T = K] = \frac{(r - q) * (C - K \frac{\partial C}{\partial K}) - \frac{dC}{dT}}{\frac{K^2}{2} \frac{\partial^2 C}{\partial K^2}} = 2 \frac{\frac{dC}{dT} + qC + (r - q)K \frac{\partial C}{\partial K}}{K^2 \frac{\partial^2 C}{\partial K^2}}$
- This is famous dupire equation that holds for all diffusive models. Simple way to satisfy LHS is to make dependance explicit. $\sigma_T = \sigma(S, T)$
- It only needs prices of vanilla options (check RHS) which are readily available. One can invert RHS to make dependence explicit in terms of implied volatility

Local Volatility: Dupire equation



SPX Implied Volatility Surface as of 24-Feb-2023



SPX Local Volatility Surface as of 24-Feb-2023

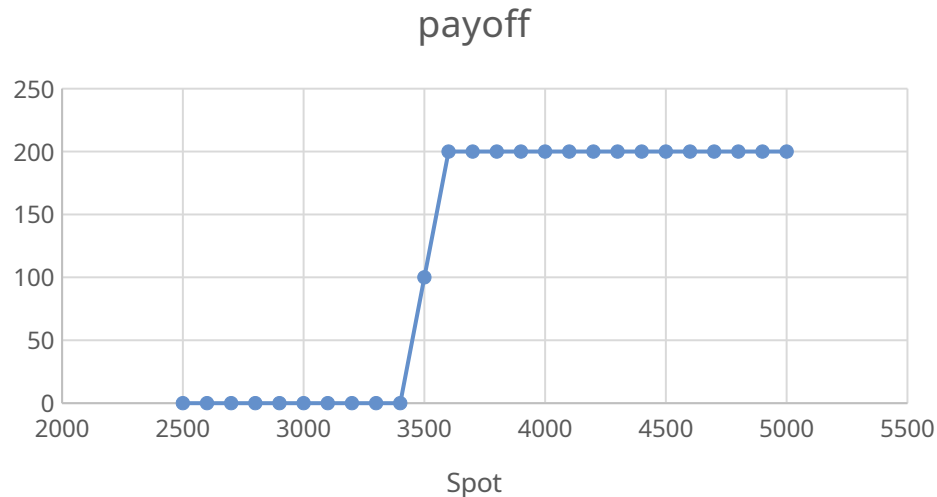
Local volatility skew is flattish and drops very rapidly

Local Volatility: Arbitrages

- $E[\sigma_T^2 | S_T = K] = 2 \frac{\frac{dC}{dT} + qC + (r-q)K \frac{\partial C}{\partial K}}{K^2 \frac{\partial^2 C}{\partial K^2}}$, this works as long as numerator and denominator are both positive.
- Denominator: $\frac{\partial^2 C}{\partial K^2} = \frac{C(\epsilon+h) + C(\epsilon-h) - 2C}{h^2}$. What is this payoff?
- Butterfly! That's why violation of condition is called butterfly arbitrage. Butterfly also give risk neutral probability density as per (2). Since its probability, it must be ≥ 0 . Hence violation of denominator are rare in practice.
- Numerator: Can be written as $e^{-qT} \frac{d}{dT} (e^{qT} C(K e^{(r-q)T}, T))$
- For this to be positive, $e^{qT} C(K e^{(r-q)T}, T)$ should increase with time (only then derivative with respect to time will have positive slope)
- This condition is known as calendar arbitrage $e^{qT_1} C(K e^{(r-q)T_1}, T_1) \leq e^{qT_2} C(K e^{(r-q)T_2}, T_2)$ for $T_1 < T_2$
- We always check for numerator and denominator for violation of calendar and butterfly arbitrage when building local volatility

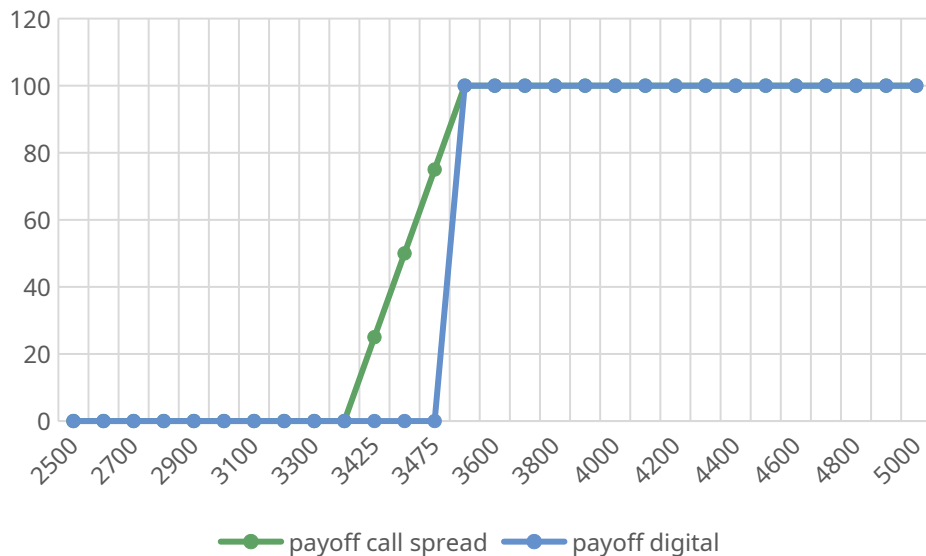
Digital Options

- Two types of digital (let's consider up and in only! There are 3 more variations)
 - Digital/ Binary Options: Payoff is 1 if $S_T \geq B$ at maturity. This is European version
 - One Touch: Payoff is 1 if $S_T \geq B$ any time during the trade life. This is American version
- How to you price this structure?
 - Using risk neutral density, find $\Pr(S_T \geq B) = 1 - \Pr(S_T < B) = 1 - \text{CDF}(B)$
 - We have already proved that $\text{CDF}(B) = 1 + \frac{\partial C}{\partial K}$
 - Hence above formula becomes $\Pr(S_T \geq B) = -\frac{\partial C}{\partial K}$
 - It can be shown by direct differentiation of BS call formula that $= -\frac{\partial C}{\partial K} = e^{-rT} N(d_2)$
 - This is closed form price of European digital option in BS world.
- Another option is to notice that $\Pr(S_T \geq B) = -\frac{\partial C}{\partial K} = \frac{C(\epsilon) - C(\epsilon+h)}{h}$
 - If h is small (tick size), we can approximate this payoff using call spread.
 - Long call at ϵ and short call at $\epsilon+h$. We buy $1/h$ calls.
 - This is called pricing by replication



- Figure shows replicating digital on SPX at barrier = 3500 (~110% spot).
- We are long call @3400 and short call @3600
- Tighter the call spread, better is the approximation.
- Such replication is used in practice but in slightly different manner
- Note that derivative of this payoff is discontinuous at barrier. Delta and Gamma can't be defined.
- This makes risk management of digital options challenging.

Digital Options



- In Practise, we use something called overhedging.
- We create a replicating portfolio (hedge) such that value of this portfolio is always greater than or equal to Barrier option
- If the barrier is at 3500, trader will book position as call spread such that $K1 = 3400$ and $K2 = 3500$.
- Price quoted to the client is price of this call spread.
- Overhedging ensures that greeks of actually booked position (call spread) are smooth
- Overhedge amount is 100 (3500-3400)

Digital Options

- Let's consider one touch (American version) now.
- Assumed zero rates and borrow for simplicity. Stock process is $dS_t = \sigma_t dW_t$
- Price of one touch $O(B) = \Pr(M_T > B)$ where M_T is maximum so far
- Price of digital/binary is $D(B) = \Pr(S_T > B)$
- If we consider price of digital when one touch hits barrier level before maturity, there is 50% chance that we end up above B by maturity.
- Hence $O(B) = 2 * D(B)$. Semi static hedging
- In BS world (log normal case where $dS_t = \sigma_t S_t dW_t$), we don't get exact relation. Rates are still 0.
- When $S=B$, once touch is worth 1 (why?).
- Digital price is $N(d_2)$.
 - Assuming 1Y maturity, 10% BS vol, $N(d_2)$ is ~0.48
- So in case of log normal dynamics, situation is different. Why?
 - There is a drift in stock process now. So it's not pure Brownian motion and we can't use reflection principle

Digital Options

- So let's again consider price of one touch when spot hits B and stock follows log normal process
- We can replicate one touch with 2 digital as before and a 1/B call option at B.
- Value of call when S=B is $SN(d1)-KN(d2)$ (rates are still zero) = $B(N(d1)-N(d2))=B(1-2N(d2)) = B(1-2D(B))$
- Hence $1 = O(B)$ when $S=B = 2D(B)+1/B*Call(B)$.
- Now let's make this more complex by adding volatility skew to BS world.
- Value of digital is $-\frac{\partial C}{\partial K} = -\frac{\partial C(B, \sigma_{imp}(B))}{\partial K} = -\frac{dC}{dK} - \frac{\partial C(B, \sigma_{imp}(B))}{\partial \sigma_{imp}(B)} \frac{\sigma_{imp}(B)}{\partial K} = D_{BS}(B) - vega * skew$
- Hence we have $O(B)=2[D_{BS}(B) - Vega_{BS} * Skew]+1/B*C(B)$
- Skew is entirely driven by your model (local vol, stoch vol etc.)